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30 September 1986
SUPERSEDING
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22 March 1966

MILITARY SPECIFICATION

JUGS, VACUUM OR INSULATED, THERMAL EFFICIENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers requirements for thermal efficient vacuum or material insulated jugs for the mass dispensing of hot or cold beverages.

1.2 Classification. The jugs shall be of the following sizes and finishes as specified (see 6.2):

Size 1 - 2 gallons
Size 2 - 3 gallons
Size 3 - 5 gallons
Size 4 - 10 gallons
Finish (1) - Satin or Butler (Manufacturer's option)
Finish (2) - Marine Corps Green 23

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7330

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	- Preservation, Methods Of
MIL-T-704	- Treatment and Painting of Materiel
MIL-R-3065	- Rubber, Fabricated Products

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-147	- Palletized Unit Loads

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Food Service Sanitation Manual, DHEW Publication No. (FDA) 78-2081

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167	- Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
A 480/A 480M	- General Requirements for Flat-Rolled, Stainless and Heat-Resisting Steel Plate, Sheet and Strip
D 3951	- Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Materials and components. Materials and components shall conform to the applicable documents as listed or required herein. Materials and components that are not covered by applicable documents or are not specifically described herein shall be of the best quality and entirely suitable for the intended purpose. All the materials shall be odorless, nonabsorbent, and nontoxic. Recycled materials may be used (see 6.4).

3.2.1 Metals. Unless otherwise specified, the metal used shall be stainless steel conforming to type 302 or 304 of ASTM A 167. The thickness of the metal for the various components shall conform to table I.

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TABLE I. Metal thickness in inches 1/

Item	Capacity in Gallons by Size			
	2 (Size 1)	3 (Size 2)	5 (Size 3)	10 (Size 4)
Inner shell:				
Bottom	0.029	0.029	0.029	0.029
Breast	0.026	0.026	0.026	0.029
Remainder	0.020	0.020	0.020	0.026
Outer shell:				
Bottom	0.026	0.026	0.026	0.029
Breast	0.026	0.026	0.026	0.045
Neck ring (if used)	0.031	0.031	0.031	0.047
Remainder	0.026	0.026	0.029	0.029
Bottom band, base (if used)	0.120	0.120	0.120	0.120
Faucet:				
Tube	0.045	0.045	0.045	0.045
Guard	0.089	0.089	0.089	0.089
Cover:				
Top	0.029	0.029	0.029	0.029
Bottom	0.025	0.025	0.025	0.026

1/ Thickness tolerances specified in ASTM A 480/A 480M shall apply.

3.3 Design. The jugs shall consist of an inner and outer shell joined in a manner to produce space for a vacuum or material insulated chamber in order to produce a thermal efficient jug. Each jug shall have an adapter capable of accommodating either a diaphragm or key type faucet. The cover shall incorporate an air vent, vent cap, and chain or other suitable attaching

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device and cam locking devices for securing the cover to the jug. The cam locking device shall be so designed that, when the cover is open, it will still remain captive to the jug. The outer shell shall be provided with reinforcing ribs, lifting handles, lugs for securing the cover, and the identification and caution plates. Figure 1 is included as a pictorial presentation of a possible jug configuration.

3.4 Construction. The jugs and components shall be manufactured in accordance with the sanitary design and construction requirements of the Food Service Sanitation Manual (see 2.2). Joints resulting from the assembly of the inner and outer shells or the attachment of the faucet tube shall be airtight with all the exposed surfaces smooth to facilitate cleaning. All interior surfaces in contact with the liquid shall be smoothly machined or polished with no crevices or pockets to trap liquids.

3.4.1 Welds. Welds shall be free from burnt holes, skips, lack of penetration, and excessively rough surfaces. Remaining flux and weld spatter shall be chipped clean and ground smooth. All joints of the inner and outer shell parts, including the union of the faucet tube to the inner shell, shall be "sanitary" type welds. All welds attaching metal components to the jug shall be smooth and shall be self-draining to prevent the accumulation of moisture and dirt.

3.4.2 Inner shell. The inner shell including the bottom, breast, and faucet tube shall be made of stainless steel. The bottom shall be sloped toward the faucet tube to afford complete removal of the contents, except for normal clingage, without tipping the jug. Figure 1 is a drawing of a typical jug meeting all the requirements of the Food Service Sanitation Manual (see 2.2).

3.4.2.1 Inner shell supports. The inner shell of the jugs shall be held in position at the bottom of non-heat-conducting supports that shall not become loose nor displaced. The supports shall prevent any lateral or vertical shifting of the inner shell with regard to its location within the outer shell.

3.4.3 Outer shell. The outer shell including the bottom, bottom bands, breast, neck ring (if used), handles, faucet guard, faucet tube adapter, and all the external components shall be made of stainless steel. The side walls of the outer shell shall be reinforced with integral continuous horizontal strengthening ribs. The number of ribs for each size shall be as specified in table II. Each rib shall be not less than 1/2 inch nor greater than 1 inch in width and shall completely encircle the outer shell. A bottom band shall be attached to the outer shell to form the foot or base of the jug. The top edges of the neck rings (if used) shall be shaped into a reinforcing bead. Two lugs

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shall be attached to the upper sides of the breast, 180 degrees apart, to engage the cam locks on the cover. The jug shall be formed with adequate provisions for the accommodation of a cover with gasket.

3.4.4 Faucet tube. The faucet tube shall be one continuous straight piece and shall have a smooth machined or polished inside diameter of not less than 13/32 inch in order to permit easy cleaning. The faucet tube shall be attached to the extreme lower radii of the inner shell bottom and outer shell as specified in 3.4.1 or by other suitable means. If welding is not used, the attachment shall be reinforced in a manner that shall prevent rupture or displacement of the faucet tube. (An adapter, with external threads to engage the wing nut or female threads of the faucet, shall be attached to the outer extremity of the faucet tube with suitable nontoxic material.) The threads of the adapter, wing nut, and key type faucet shall conform to figure 1 to provide for interchangeability of the faucets.

3.4.5 Faucet. The faucet may be either the key or diaphragm type faucet. Each faucet shall be so constructed to ensure an airtight attachment to the jug. The handle shall be suitable for the purpose and shall provide positive on-and-off type operation. There shall be a minimum of moving parts employed in the liquid contact zone. The faucet shall be constructed to permit easy dismantling and reassembly without the use of tools. All corners, crevices, depressions, and threads shall be constructed to permit simple yet thorough cleaning of both interior and exterior surfaces. Parts not made of stainless steel shall be entirely suitable for the purpose and shall have a finish as specified in 3.8. Each type of faucet shall contain a nonabsorbent washer made of fiber or other suitable nontoxic material. Any anti-splash configurations within the spout shall not create difficult-to-clean areas. All components shall be able to withstand maximum commercial cleaning and sanitizing temperatures.

3.4.5.1 Key type. All food contact surfaces of the key type of faucet shall be constructed from stainless steel. The faucet bonnet and stem assembly of the faucet shall be made of chromed brass or stainless steel. To facilitate cleaning, the liquid passage holes shall be designed, machined, and polished to eliminate "dead ends" or sharp angles.

3.4.5.2 Diaphragm type. All food contact surfaces of the diaphragm type of faucet shall be constructed of stainless steel. The faucet bonnet and stem assembly of the faucet shall be made of chromed brass or stainless steel. The handle shall be equipped with a simple device to lock itself in the closed position. The nozzle shall have a minimum opening of 1/2 inch.

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3.4.6 Faucet guard. A faucet guard, with suitable radii, shall be securely attached by welding to the outer shell of the jug. The faucet guard shall surround and protect the diaphragm type or key type of faucet from inadvertent operation and bumping. The guard shall permit access for easy removal of the diaphragm type or key type of faucet.

3.4.7 Cover. The top and bottom of the cover shall be joined together by welding or soldering. The cover shall be shaped to prevent the accumulation of moisture. An air vent tube, having a smooth inside diameter of not less than 1/2 inch, shall extend through and sufficiently beyond the cover to prevent the introduction of contaminants from surface drainage. The vent tube shall be equipped with a stainless steel or molded nylon vent cap with "sanitary" threads. The vent cap, when closed, shall be airtight and leakproof and, when partially opened but not removed, shall vent the jug. If solder is used instead of welding in the assembly of the cover parts, it shall be of a nontoxic composition (see 3.2). A suitable stainless steel chain, or other suitable attaching device, shall be provided to attach the vent cap to cover.

3.4.7.1 Cover locking device and gasket. Each cover shall contain two cam-locking devices, attached 180 degrees apart, on the periphery of the cover top. These devices shall engage the corresponding lugs welded to the neck of the outer shell in a manner that shall compress the cover gasket to create an airtight, leak-proof seal. The cover gasket shall be a replaceable, one-piece unit constructed of material conforming to MIL-R-3065.

3.4.8 Handles. The sizes 1, 2, and 3 jug shall be provided with two rigid handles, each with or without brackets. Each size 4 jug shall be provided with two drop handles, each complete with brackets. The rigid handles shall be located diametrically opposite each other on the breasts or upper side walls of the jug. The drop handles shall be located diametrically opposite each other on the upper side walls only.

3.4.9 Space between inner and outer shells. The thermal efficiency of the jug shall result from the space between the inner and outer shells being filled with material insulation or by creation of a vacuum.

3.4.9.1 Material insulation. If material insulation is used, it shall be nonhygroscopic, shall not cause corrosion, nor in any other way be deleterious to the inner and outer shells. The material insulation shall be nonsettling, nonshifting, and of a composition and quality suitable for use as insulation material for the jug. The material insulation shall be bonded to the outer surface of the inner shell and to the inner surface of the outer shell.

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3.4.9.2 Vacuum insulation. If jug is to be insulated by a vacuum, air shall be exhausted from the space between the inner and outer shells. The space shall be permanently sealed and have a vacuum capable of raising a column of mercury in a 1/8-inch inside diameter glass tube to a minimum height of 28 inches. A dial-type gage calibrated to express the degree of vacuum in terms of raising a column of mercury in a tube of the specified diameter may be used instead of the actual mercury and tube.

3.5 Capacities and dimensions. The jugs shall have a minimum liquid service capacity of 2 (size 1), 3 (size 2), 5 (size 3), and 10 (size 4) gallons, respectively, and the dimensions shall conform to table II.

TABLE II. Jug dimensions in inches

	Capacity in Gallons by Size				
Location	2 (size 1)	3 (size 2)	5 (size 3)	10 (size 4)	Tolerance
Height	19 3/4	20 3/4	26 1/2	33 1/2	Maximum
Diameter of body	13	13	13	16	Maximum
Diameter of body including guard	17	17	17	20	Maximum
Distance between bottom of jug and bottom of faucet	3 3/16	3 3/16	4 7/16	4 7/16	Minimum
Diameter of opening (neck into container)	5 1/2	5 1/2	5 1/2	5 1/2	±1/2 inch
Number of ribs	3	3	5	7	Minimum

3.6 Performance.

3.6.1 Thermal efficiency. When inspected as specified in 4.6.1, the maximum temperature loss of the contents of the jugs shall not exceed the following:

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Size	Maximum temperature loss (degrees F)
1	80
2	70
3	65
4	55

3.6.2 Durability. The assembled jug, after having been repeatedly dropped as specified in 4.6.2, shall conform to the requirements for buoyancy in 3.6.5. All components shall remain perfectly intact and shall show no indication of separation. Upon completion of the durability inspection, the assembled jug shall conform to the requirements of the strength of handle attachment inspection.

3.6.3 Random free drop. The assembled jug, after having been dropped as specified in 4.6.3, shall conform to the requirements for durability in 3.6.2. This inspection shall be performed in sequence with the durability inspection.

3.6.4 Strength of handle attachment. Each handle, when inspected as specified in 4.6.4, shall not deform, rupture, crack, or separate and the jug shall conform to the requirements for thermal efficiency in 3.6.1.

3.6.5 Buoyancy. The assembled jug shall not sink or increase in weight, when inspected as specified in 4.6.5.

3.6.6 Weight. The assembled jug, when weighed as specified in 4.6.6, shall conform to table IV:

Size	Weight, maximum (pounds)
1	15 1/2
2	20
3	25
4	40

3.7 Markings.

3.7.1 Data name plate. The jugs shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data name plate shall be made of minimum 20 gage corrosion-resisting metal, and shall be attached to the jugs by rivets, screws, or by welding, in such a manner as to meet the applicable National Sanitation

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Foundation requirements for this equipment. The plate shall bear the following information, which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (see 6.2)

Each plate shall be placed so as to be readily visible to the operator during normal operation and use, and so as to not adversely affect the life or utility of the jug.

3.7.2 Caution. A stainless steel caution plate shall be attached to the front or back of the jug in a conspicuous location. The inscription in permanent black letters shall be as follows:

DO NOT USE FOR MILK OR MILK PRODUCTS

3.8 Finish. The interior surface of the inner shell, including the neck, and the exterior surfaces of the outer shell, including the cover, all cover parts and fittings, the handles, and the faucet guard, shall have a satin or butler finish (Finish (1)). All chromium-plated fittings shall be polished to a commercial No. 7 Finish. When specified, the outer shell, including the cover, shall be finished in Marine Corps Green semigloss No. 23 (Finish (2)) conforming to MIL-T-704, type A. A satin or butler finish is not required (as an underfinish) when the Marine Corps Green is specified. The vent cap, faucet, and the identification and caution plates shall not be painted.

3.9 Workmanship. The assembled jugs, including all components, shall be constructed and finished so that they shall be free from any defect that may affect proper functioning. Particular attention shall be given to neatness and thoroughness of soldering, marking, welding, plating, painting, and freedom of parts from burrs and sharp edges. The lugs shall conform to the quality and grade of product established by this document. The occurrence of defects shall not exceed the acceptance criteria established herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Testing conditions. All testing required by 4.5.4 of this document shall be performed at an atmospheric pressure of 28 to 30 inches of mercury and at a temperature of 77° ±18°F.

4.4 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects specified in 4.5.2 and 4.5.3 and tested for the characteristics specified in 4.5.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.2 End item visual examination. The jugs shall be examined for the defects listed in table III. The lot size shall be expressed in units of assembled jugs of one size made under essentially the same conditions and from the same materials and components. The sample unit shall be one assembled jug of one size. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 15.0 for minor defects.

TABLE III. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not a satin or butler finish (where applicable)	101	
	Any chromium-plated surface not a polished commercial No. 7, high luster finish		201
	Outer shell and cover not a green semigloss paint (when applicable)		202
	Any paint on vent cap, faucet, exposed faucet tube, identification or caution plate		203
	Any interior surface having roughness over 1 square inch	102	
	Any deep scratch or tool mark	103	
Construction and workmanship	Design characteristics not as specified	104	
	Inner or outer shell or any component malformed, fractured, dented, buckled, or bent	105	
	Any sharp edges, burrs, slivers, or rough edges	106	
	Inner shell bottom not sloped toward the faucet tube	107	

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TABLE III. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship - continued	Outer shell not reinforced with continuous horizontal strengthening ribs (where applicable)	108	
	Inner shell not securely supported at bottom	109	
	Bottom band not attached to the outer shell or does not form a foot or base for jug	110	
	Top edge of neck ring (if used) not shaped into reinforcing beads	111	
	Failure of the cap to be captive when cover is open	112	
	Materials not as specified (where applicable)	113	
	NOTE: Ripples or buckles on bottom of jug caused by air evacuation from between the inner and outer shells shall not be classified as a defect		
Cover and locking device	Lugs not attached as specified or do not engage the cam locks on cover	114	
Diaphragm type or key type of faucet and tube	Not attached to lower radii of inner shell bottom as specified	115	
	Faucet guard missing or not securely attached		204
Handle	Not specified type		205
Gasket	Not one piece or not molded to fit snugly	116	
	Missing or not of the replaceable type	117	
Chain	Missing		206

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TABLE III. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Welding or soldering	Omitted, cracked, burned through, or not specified type of weld	118	
	Excess flux or spatter not removed	119	
	Excess of material not ground smooth	120	
Data name plate	Omitted or not as specified		207
	Information incomplete or not legible		208
	Not located so as to be readily visible to operator		209
Caution marking	Missing, illegible, or not applied as specified		210

4.5.3 End item dimensional examination. Examination shall be made of the jugs for compliance with the dimensions specified in table II and the minimum nozzle opening specified in 3.4.5.2. Any deviation from specified requirements shall constitute a defect. The lot size shall be expressed in units of assembled jugs of one size made under essentially the same materials and components. The sample unit shall be one assembled jug of one size. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.5.4 End item testing. When a first article is not required, the first produced jug shall be subjected to all tests specified in table IV. Failure of any test shall be cause for rejection of the first produced lot. Testing of the completely fabricated jug shall be performed as specified in table IV.

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TABLE IV. Test methods for jugs

Characteristic	Requirement para.	Test method	Number of determinations per sample unit	Results reported as	Sample size	Acceptance criteria
Thermal efficiency	3.6.1	4.6.1	1	To nearest degree	Inspection level S-4	Acceptance number zero, rejection number 1
Durability 1/	3.6.2	4.6.2	15,000	Pass or fail	One out of each 500 or fraction thereof	Acceptance number zero, rejection number 1
Random free drop 1/	3.6.3	4.6.3	5	Pass or fail	One out of each 500 or fraction thereof	Acceptance number zero, rejection number 1
Strength of handle attachment 1/	3.6.4	4.6.4	1	Pass or fail	Inspection level S-2	Acceptance number zero, rejection number 1
Buoyancy	3.6.5	4.6.5	1	Pass or fail	Inspection level S-2	Acceptance number zero, rejection number 1
Weight	3.6.6	4.6.6	1	To the nearest 0.1 pound	Inspection level S-3	Acceptance number zero, rejection number 1

1/ These inspections shall be performed in sequence: first durability, next random free drop, and then strength of handle attachment.

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4.5.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers containing assembled jugs of one size. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Contents	Contains jugs of more than one size
Materials	Any nonconforming component; component missing, damaged, or otherwise defective
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or tape. Bulging or distortion of containers
Weight of container	Gross weight exceeds requirements
Marking	Omitted, incorrect, incomplete, illegible; of improper size and location, sequence, or method of application; markings not the same on the interior or exterior container

4.5.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.6 Methods of inspection.

4.6.1 Thermal efficiency. The jug, including cover and faucet, shall be tempered or preheated with water at a temperature of 204° to 207°F for 5 minutes. The jug shall then be emptied and refilled with water at a temperature of 204° to 207°F to the rated capacity, and the cover secured. The decrease in temperature of the contents after 12 hours shall not exceed the limit specified in 3.6.1. A still atmospheric temperature of between 70° and 76° F shall be maintained for the duration of the test.

4.6.2 Durability. The jug shall be filled with water to the rated capacity and placed in a suitable testing machine, which will repetitively lift the jug to a height of 3 inches and drop it onto a firm steel base at the rate of 1,500 drops per hour, until each jug under test has been dropped a total of 15,000 times. Upon completion of the aforementioned, the jug shall be examined. All the components shall remain perfectly intact and shall show no indication of separation.

4.6.3 Random free drop. The sizes 1, 2, and 3 jug, as applicable, shall be filled to the rated capacity and tested in a suitable machine that will permit the jug to fall freely through a trap-door device from a height of not less than 37 inches onto a firm steel base. The jug under test shall be placed in the identical position for each drop and shall be dropped a total of five times as follows: The jug shall be level for the first drop; thereafter, a 3-inch wooden cube shall be placed under the bottom rim of the jug to vary the angle of each drop. For each successive drop, the cube of wood shall be positioned under the bottom rim 90 degrees in a clockwise direction from the previous drop. No restraining or guide lines of any kind shall be used. The size 4 jug shall be similarly tested except the height of drop shall be 19 inches.

4.6.4 Strength of handle attachment. Each handle of the sizes 1, 2, or 3 jug, as applicable, shall be tested separately by suspending from it a weight of 100 pounds for 5 minutes. The size 4 jug shall be tested as above except the suspended weight shall be 200 pounds. Each jug shall be rigidly secured with the required weight suspended from the hand grip of the handle in a manner that will develop the greatest stress. The weight shall be uniformly distributed over the hand grip and shall hang free. Upon completion of the durability, random free drop, and strength of handle attachment tests, the jug shall be subjected and conform to the thermal efficiency and buoyancy tests.

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4.6.5 Buoyancy. The empty jug shall be weighed. The jug shall then be filled with water at a temperature of 70° ±5°F to rated capacity and submerged in a tank of water having a temperature of 70° ±5°F. After having been submerged for 1 hour, if the jug is not capable of floating, the sample shall be rejected. The jug shall then be emptied and weighed again. The sample shall be rejected if there is any increase in weight.

4.6.6 Weight. The weight of the assembled jugs shall be determined on a scale capable of weighing to the nearest 0.1 pound. The weight of the jug shall be as specified in 3.6.6.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each jug shall be individually unit packed and cushioned, in accordance with Method III of MIL-P-116, in a container conforming to type CF or SF, class domestic of PPP-B-636.

5.1.2 Commercial. The jugs shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. The jugs, preserved as specified in 5.1, shall be packed in overseas exterior type containers conforming to overseas type, style optional, grade A, type 2 load of PPP-B-601. Closure and strapping of the container shall be in accordance with PPP-B-601.

5.2.2 Level B packing. The jugs, preserved as specified in 5.1, shall be packed in domestic exterior type containers conforming to class domestic, grade 275 of PPP-B-636. Closure and strapping of the containers shall be in accordance with the appendix of PPP-B-636. Gross weight of the packed shipping containers shall not exceed the weight limitations of PPP-B-636.

5.2.3 Commercial packing. The jugs, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), shipping containers shall be palletized in accordance with MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The jugs are intended to be used to keep liquids cold or hot during military operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size and finish required (see 1.2).
c. Government approved manual number to be included in data name plate (see 3.7.1).
d. When a first article is required (see 4.4).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical, as long as it meets the requirements of this document (see 3.2).

6.5 Subject term (key word) listing.

Containers
Jugs

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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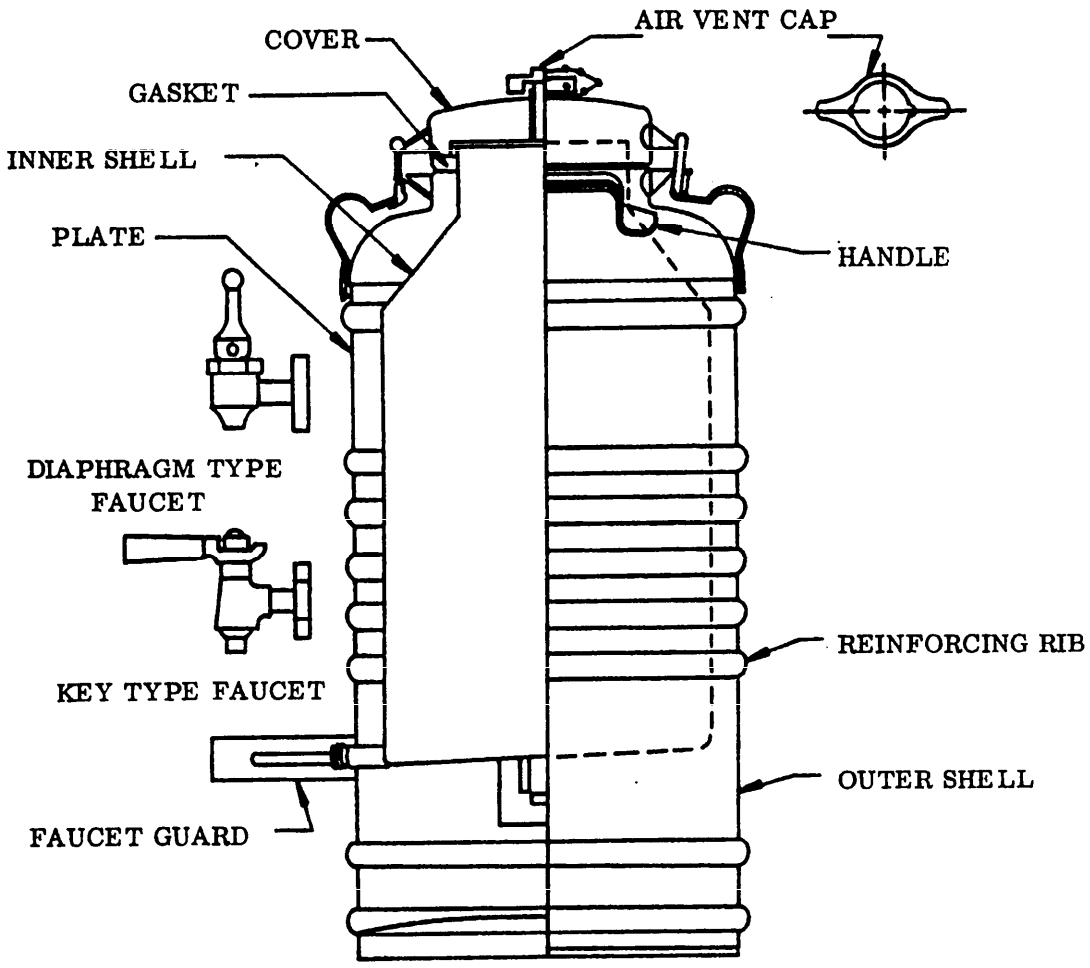
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Preparing activity:

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NOTE: Threads shall be 7/8-14UNF-2B conforming to NBS Handbook H28

This drawing is a pictorial presentation only.

Figure 1. Jug Configuration

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-J-23049C	Jugs, Vacuum or Insulated, Thermal Efficient

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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